

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 2.4: SURFACE AREA AND VOLUME OF SOLIDS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have spent this sub-strand investigating the puzzle from the Kisumu water tank factory: two tanks that each hold 5,000 litres of water, yet the frustum tank uses less metal sheet than the cylinder-hemisphere tank. Use your notes, your formulae, your Comparison Table and your cumulative model to answer each question below. Write in full sentences, show your working where calculations are needed, and use the correct mathematical words (surface area, volume, slant height, net, composite solid, frustum). Use the data in the table in Section 1 whenever a question refers to it. Metal sheet costs KSh 850 per square metre.

Section 1: Reading the evidence from the Kisumu factory

The table below shows the results you calculated for the two tanks made at the Kisumu factory for households in Siaya County.

Tan k desi gn	Volu me of wate r held	Tota l surf ace area	Meta l shee t cost at KSh 850 per m ²
Cylin der with hemi sphe rical lid	appr ox. 5,00 0 litres	appr ox. 16.0 9 m ²	appr ox. KSh 13,6 77

(a) The frustum tank needs less metal sheet. The difference in surface area is about $16.09 - 7.03 = 9.06 \text{ m}^2$. In cost, the difference is about KSh 13,677 – KSh 5,975 = about KSh 7,700 saved on metal sheet for each tank. (b) The students saw that both tanks hold the same amount of water, so they have the same volume (about 5,000 litres). They expected that tanks holding the same amount would need the same amount of metal. But the frustum has a much smaller surface area, so it needs far less sheet. This showed that volume and surface area are not linked in the way they first thought.

Frustum	approx. 5,000 litres	approx. 7.03 m ²	approx. KSh 5,975
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(a) Which tank needs less metal sheet? By roughly how many square metres, and by roughly how much money, does it differ from the other tank?

(b) Explain in your own words why the students at the factory were puzzled. Use the words volume and surface area in your answer.

Section 2: Why surface area and volume are independent

(a) What does 'surface area' of a solid tell a manufacturer, and what does 'volume' tell a manufacturer? Give one example of each from the water tank factory.

(b) Use your work on the two tanks to explain what it means to say that surface area and volume are independent properties of a solid.

(c) A cone and a cylinder have the same base radius and the same vertical height. Compare their volumes and compare their total surface areas. What does this comparison show about how shape affects each quantity?

(a) Surface area tells how much material is needed to build or cover the solid, for example the metal sheet needed to make a tank. Volume tells how much the solid can hold, for example the 5,000 litres of water in the tank. (b) Independent means that one quantity does not decide the other. The cylinder-hemisphere tank and the frustum tank have the same volume (about 5,000 litres) but different surface areas (about 16.09 m² and 7.03 m²). So you can change the shape to reduce the surface area without changing how much water the tank holds. (c) The cone's volume is one-third of the cylinder's volume ($V = \frac{1}{3}\pi r^2 h$ compared with $V = \pi r^2 h$). The cone also has a smaller total surface area than the cylinder. So shape changes both quantities, but the amounts do not change in the same way, which again shows that shape affects surface area and volume differently.

Section 3: Explaining the result using formulae and nets

(a) The total surface area of the frustum tank is found using $SA = \pi(R^2 + r^2) + \pi(R + r)l$. Name the part of the tank that each of the three terms πR^2, πr^2 and $\pi(R + r)l$ represents. (b) When you found the surface area of the cylinder-hemisphere tank, you worked with a composite solid. Describe the three steps of the strategy you used and explain what happens at the joint between the cylinder and the hemisphere. (c) Using your answers above and the table in Section 1, explain why the frustum tank has the smaller surface area for the same volume.	(a) πR^2 is the area of the larger circular base, πr^2 is the area of the smaller circular top, and $\pi(R + r)l$ is the area of the curved sloping side (the lateral surface), where l is the slant height. (b) Step 1: identify the simple solids that make up the composite solid (a cylinder and a hemisphere). Step 2: subtract the hidden faces at the joint, because the circle where the pieces meet is not on the outside of the tank. Step 3: add the areas of all the exposed surfaces. (c) The cylinder-hemisphere tank has a tall curved cylinder wall plus a curved hemispherical lid, which together expose about 16.09 m² of surface. The frustum has only a base, a narrower top and one sloping side, which expose about 7.03 m². Its sloping side, whose size depends on the slant height and the two radii, lets the frustum hold the same 5,000 litres with much less outer surface, so less metal sheet is needed.

Section 4: Engineer's recommendation for Nyakach Secondary School	
You are a junior engineer at a Kenyan water tank manufacturer. Nyakach Secondary School in Siaya County needs a tank that holds 5,000 litres. Metal sheet costs KSh 850 per square metre. (a) Using the table in Section 1, show how the metal sheet cost of the frustum tank is calculated from its surface area. (b) Which of the two designs in the table would you recommend to the school? Justify your choice with at least two pieces of mathematical evidence. (c) Name one other thing, apart from metal sheet cost, that an engineer or the school might want to consider before the final choice is made.	(a) Cost = surface area $\times$ price per square metre = $7.03 \times 850 \approx$ KSh 5,975. (b) I would recommend the frustum tank. Evidence 1: both tanks hold about 5,000 litres, so the frustum meets the school's capacity requirement just as well as the cylinder-hemisphere tank. Evidence 2: the frustum's surface area is about 7.03 m² compared with about 16.09 m², so its metal sheet costs about KSh 5,975 compared with about KSh 13,677, a saving of about KSh 7,700. Using less metal also means less material to transport and use up. (c) Any sensible answer, for example: how stable the tank is when standing on the ground, how easy it is to make, clean or fit a lid, or how much space is available at the school.

Section 5: Using this mathematics in Kenya

(a) Describe one real situation in Kenya, other than the Kisumu water tanks, where someone needs to know the surface area of a solid, and one where someone needs to know its volume. Name the solid or solids involved. (b) Choose one career (for example architecture, agriculture, manufacturing or water engineering) and explain how the idea that shape affects surface area and volume independently helps a person in that career to make good decisions.	(a) Surface area: a roofing contractor in Nairobi calculating the metal sheeting needed for a conical roof turret, or the sheet metal needed to make a frustum-shaped grain store in Kitale. Volume: a farmer in the Rift Valley working out how much maize a frustum-shaped grain store (mapunja) can hold, or a tea company in Kericho working out how much tea a canister made from a cylinder and a cone can hold. (b) A water engineer must supply a fixed amount of water, such as 5,000 litres, to a school or village. Because surface area and volume are independent, the engineer can compare different shapes that all hold 5,000 litres, calculate the surface area of each, and choose the shape that needs the least metal sheet. This lowers the cost for rural households and uses resources responsibly, which supports water conservation.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses the table values and the Kisumu tank data accurately and refers to them in every answer, including correct differences and costs.	Uses most of the table values correctly, with minor slips in differences or costs.	Uses little of the data or quotes values incorrectly.
Understanding that surface area and volume are independent	Clearly explains with the two tanks and the cone and cylinder comparison that equal volume can come with different surface areas, and that shape affects each quantity differently.	States that surface area and volume are independent and supports it with one example.	Confuses surface area with volume or says a larger volume always means a larger surface area.
Mathematical reasoning with formulae and strategies	Correctly identifies each term in the frustum formula, describes the three-step composite solid strategy including hidden faces, and links these to the result.	Identifies most terms and steps, with small gaps in explaining hidden faces or slant height.	Names formulae or steps incompletely or incorrectly.
Justified recommendation and Kenyan applications	Gives a clear recommendation supported by at least two pieces of evidence, and gives accurate, relevant Kenyan examples and a career link.	Gives a recommendation with some evidence and at least one suitable Kenyan example.	Gives a recommendation without evidence or gives examples that are unclear or unrelated.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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